

# Keugene Lee

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## SUMMARY

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**Full-stack and applied-AI engineer** who ships AI products end to end. Built a job-search agent that indexes 20,000+ career boards and files applications through a browser extension, a personal AI agent that keeps **45-minute tasks alive on 300-second serverless functions** and ships to iOS and Android, and a retrieval agent that answers equipment questions with cited manual pages. Mathematics major. Works across TypeScript, Python, Next.js, Postgres, Docker, and CI/CD.

## EXPERIENCE

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### Understudy | [getunderstudy.vercel.app](https://getunderstudy.vercel.app)

September 2026 – Present

- Built an AI job-search agent that finds matching roles, tailors a resume, and fills out the application, indexing open roles from **20,431 validated career boards across 7 applicant tracking systems** (Greenhouse, Lever, Ashby, Workday, Workable, SmartRecruiters, Recrutee) into one normalized Postgres schema.
- Held model cost to **\$0.005 per application** (\$0.29 across 600+ calls) by running GLM-5.3-Flash on **OpenRouter** at \$0.15/\$0.50 per 1M tokens instead of Claude Opus 5 at \$5/\$25 — about **40x cheaper** — behind a two-stage pipeline that filters on structured fields before calling a model.
- Automated authenticated applications end to end through a Chrome MV3 extension and headless Playwright, creating per-employer ATS accounts and sealing generated credentials with **AES-256-GCM** (random IV, auth tag, server-side key) in Postgres.

### Tarrow | [tarrow.vercel.app](https://tarrow.vercel.app) | iOS & Android

September 2026 – Present

- Built a personal AI agent that plans and runs multi-step tasks — web research, Maps and events, calendar cards, message drafts, and approval-gated recurring routines — shipping **10,700 lines of TypeScript across 44 REST endpoints** to web, iOS, and Android from one Next.js codebase via Capacitor.
- Kept **45-minute agent runs alive on 300-second serverless functions** by making every run a Postgres row advanced by chained, lease-based worker invocations with crash recovery, streaming progress over **SSE** and running parallel sub-agents as child runs that wake the parent on completion.
- Routed inference across **GLM-5.3-Flash, DeepSeek, and Groq** with rate-limit-aware failover and per-call cost tracking, and verified subscriptions **server-side against Stripe, StoreKit 2, and Google Play** rather than a third-party service, with signed iOS and Android builds produced in GitHub Actions.

### Prox | [prox-challenge-sooty.vercel.app](https://prox-challenge-sooty.vercel.app) | Founding Engineer Technical Challenge

April 2026

- Built a retrieval-augmented agent that answers technical questions about a machine and **grounds every answer in cited pages** of its 102-page manual, indexing per-page text plus vision-model captions behind five tools on a hand-rolled Messages API loop.
- Cut per-turn latency **50%, 24s to 12s**, and brought the deploy under Vercel's 250MB function limit by replacing the Agent SDK's bundled CLI binary with a direct Messages API implementation.

## MATHEMATICS

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### Independent Study & Competition

Summer 2025

- Competed at the **International Mathematics Competition** in Sofia, Bulgaria (August 2025).
- Directed Reading Program in Algebraic Geometry, mentored by a Stony Brook graduate student (June–July 2025).
- Nesin Mathematics Village (Şirince, Turkey): advanced courses in Graph Theory, Number Theory, and Quadratic Fields (July 2025).
- Coursework: Introduction to Analysis, Advanced Linear Algebra, Differential Equations, Multivariable Calculus.

## EDUCATION

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### Stony Brook University

2024 – Nov. 2025

*B.S. Mathematics coursework, 3.6 GPA — withdrew to build software full-time*

## TECHNICAL SKILLS

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**AI & LLM:** Claude API, OpenAI API, function calling, grounded retrieval, multi-provider routing, evals

**Languages & Data:** TypeScript, JavaScript, Python, SQL, PostgreSQL (Neon, Supabase), REST APIs, SSE

**Platforms & Tools:** React, Next.js, Node.js, Capacitor, Chrome Extensions, Playwright, CI/CD, Docker, OAuth 2.0